



AWS Audit Report

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Report Date	2026-05-12
AWS Account ID	798094440826
Number of tests	27
Passed	15 (55.6%)
Failed	12 (44.4%)

Disclaimers

- This report was generated using logic from the AWS Audit Playbook (<https://github.com/ajdehn/AWS-Audit-Playbook>).
- Evidence used to conduct the audit was gathered directly from boto3 (AWS software development kit 'SDK' for Python).
- Evidence used to produce this audit was gathered as of the 'Report Date' above. Configurations may have changed since this report was generated.

Test Summary

Test Description	Results	Risk Level	Comments
IAM users with an active console password have MFA enabled.	Fail	High	Exceptions Noted. 4 IAM user(s) do not have MFA enabled.
IAM user access keys are rotated at least every 90 days.	Fail	High	Exceptions Noted. 1 IAM key(s) are over 90 days old.
GuardDuty is enabled for all in-scope regions.	Fail	High	Exceptions Noted. GuardDuty is not enabled for 1 in-scope region(s).
EBS volumes are encrypted at rest.	Fail	Medium	Exceptions Noted. 1 EBS volume(s) are not encrypted.
CloudTrail trails have log file validation enabled.	Fail	Medium	Exceptions Noted. 1 trail(s) do not have log file validation enabled.
S3 buckets must have required tags applied and tag values must not be empty.	Fail	Low	Exceptions Noted. 2 bucket(s) missing required tags or have empty values.
RDS backups are retained for at least 14 days.	Fail	Low	Exceptions Noted. 1 RDS instance(s) do not retain backups for at least 14 days.
EBS volumes must have required tags applied and tag values must not be empty.	Fail	Low	Exceptions Noted. 1 EBS volume(s) are missing required tags or have empty values.
EC2 instances must have required tags applied and tag values must not be empty.	Fail	Low	Exceptions Noted. 1 EC2 instance(s) are missing required tags or have empty values.
EC2 security groups have required tags applied and tag values are not be empty.	Fail	Low	Exceptions Noted. 4 security group(s) are missing required tags or have empty values.
S3 buckets are configured to deny unencrypted data in-transit.	Fail	Informational	Exceptions Noted. 5 S3 bucket(s) do not enforce secure transport (HTTPS).
EBS volumes must have default encryption enabled in each region.	Fail	Informational	Exceptions Noted. 2 region(s) do not have EBS default encryption enabled.
Root account has MFA enabled.	Pass	High	

Root account does not have any active access keys.	Pass	High	
S3 buckets are configured to block public access.	Pass	High	
RDS instances are not publicly accessible.	Pass	High	
At least one multi-region CloudTrail trail has logging enabled.	Pass	High	
CloudTrail S3 buckets are configured to block public access.	Pass	High	
CloudTrail logging has not been stopped in the last 365 days.	Pass	High	
IAM passwords must comply with the organizations password complexity requirements.	Pass	Medium	
S3 buckets are encrypted at rest.	Pass	Medium	
RDS instances are encrypted at rest.	Pass	Medium	
RDS instances have deletion protection enabled at the cluster or instance level.	Pass	Medium	
WAF is enabled on Application Load Balancers and API Gateways.	Pass	Medium	
RDS instances have automatic minor version upgrades enabled.	Pass	Low	
RDS instances must have required tags applied and tag values must not be empty.	Pass	Low	
Lambda functions must have required tags applied and tag values must not be empty.	Pass	Low	

Test ID	iam_users_mfa
Test Description	IAM users with an active console password have MFA enabled.
Risk Rating	High
Test Procedures	1. Obtained a list of IAM users by calling the list_users() boto3 command. 2. Saved the list of IAM users: iam/users.json. 3. For each IAM user, obtained the login profile information by calling the get_login_profile() boto3 command. 4. For each IAM user, saved the login profile: iam/users/[user_name]/login_profile.json. 5. Saved the login profile for each user in the audit evidence folder (iam/users/[user_name]/login_profile.json). 6. For each IAM user with a login profile, obtained the MFA device information by calling the list_mfa_devices() boto3 command. 7. For each IAM user with a login profile, saved the MFA device information: iam/users/[user_name]/mfa_devices.json] 8. For each IAM user with a login profile, inspected mfa_devices.json to determine if at least one MFA device is registered.
Conclusion	Fail
Comments	Exceptions Noted. 4 IAM user(s) do not have MFA enabled.

IAM User Name	Result	Comments
goose	Fail	No MFA device enabled for this user.
iceman	Fail	No MFA device enabled for this user.
maverick	Fail	No MFA device enabled for this user.
slider	Fail	No MFA device enabled for this user.
audit-user	Pass	
itauditguy	Pass	
student1	Pass	

Test ID	iam_user_access_key_age
Test Description	IAM user access keys are rotated at least every 90 days.
Risk Rating	High
Test Procedures	1. Obtained a list of IAM users by calling the list_users() boto3 command. 2. Saved the list of IAM users: iam/users.json. 3. For each IAM user, obtained access key metadata by calling the list_access_keys() boto3 command. 4. For each IAM user, saved access key metadata: iam/users/[user_name]/access_keys.json 5. Inspected the 'AccessKeyMetadata' for each user to determine if they comply with the test attribute(s) below.
Test Attributes	• 'CREATE_DATE <= 90 days ago (for keys with an 'ACTIVE' status).
Conclusion	Fail
Comments	Exceptions Noted. 1 IAM key(s) are over 90 days old.

User	Access Key ID	Result	Comments
itauditguy	AKIA3TURCXF5GCAELVHX	Fail	Key is 1205 days old.
audit-user	AKIA3TURCXF5OWBHPICR	Pass	
student1	AKIA3TURCXF5K4HRB55Z	Pass	

Test ID	guardduty_enabled
Test Description	GuardDuty is enabled for all in-scope regions.
Risk Rating	High
Test Procedures	1. For each in-scope region, obtained a list of GuardDuty detectors by calling the list_detectors() boto3 command. 2. For each in-scope region, saved the list of detector IDs: guardduty/[region]/detectors.json. 3. For each detector ID, obtained detector configuration by calling the get_detector() boto3 command. 4. For each detector ID, saved the detector configuration: guardduty/[region]/[detector_id]/config.json. 5. For each detector ID, inspected the detector configuration to determine whether 'Status' is set to 'ENABLED'.
Conclusion	Fail
Comments	Exceptions Noted. GuardDuty is not enabled for 1 in-scope region(s).

Region	Result	Comments
us-east-2	Fail	Detector(s) found but none are enabled.
us-east-1	Pass	

Test ID	ebs_volume_encryption
Test Description	EBS volumes are encrypted at rest.
Risk Rating	Medium
Test Procedures	1. For each in-scope region, obtained a list of EBS volumes by calling describe_volumes() boto3 command. 2. For each in-scope region, saved the list of EBS volumes: ec2/[region_name]/volumes.json. 3. For each EBS volume, inspected the 'Encrypted' attribute to determine it is set to 'true'.
Conclusion	Fail
Comments	Exceptions Noted. 1 EBS volume(s) are not encrypted.

Region	Volume ID	Result	Comments
us-east-1	vol-0fad5b6c1c9198060	Fail	EBS volume is not encrypted.

Test ID	cloudtrail_log_file_validation
Test Description	CloudTrail trails have log file validation enabled.
Risk Rating	Medium
Test Procedures	1. Obtained CloudTrail trails using the describe_trails() boto3 command. 2. Saved the trail configuration in the audit evidence folder (cloudtrail/trails.json). 3. Inspected each trail's configuration to determine if 'LogFileValidationEnabled' was set to True for all trails.
Conclusion	Fail
Comments	Exceptions Noted. 1 trail(s) do not have log file validation enabled.

Trail Name	Result	Comments
management-events	Fail	Log file validation is disabled.

Test ID	s3_tags
Test Description	S3 buckets must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. Obtained a list of S3 buckets by calling the list_buckets() boto3 command. 2. Saved the list of buckets: s3/buckets.json. 3. For each bucket, obtained its tags by calling the get_bucket_tagging() boto3 command. 4. For each bucket, saved the tags: s3/buckets/[bucket_name]/tags.json. 5. For each bucket, inspected the tags to determine if the following tag keys exist and have non-empty values: ['owner', 'description', 'classification']
Conclusion	Fail
Comments	Exceptions Noted. 2 bucket(s) missing required tags or have empty values.

Bucket Name	Result	Comments
aws-cloudtrail-logs-798094440826-aa81e64c	Fail	Missing tags: ['classification'].
config-bucket-798094440826	Fail	Tags not found on this bucket.
bad-bucket-54321	Pass	
itauditguy	Pass	
taylor-swift-secret-vault	Pass	

Test ID	rds_backup_retention
Test Description	RDS backups are retained for at least 14 days.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained a list of RDS instances by calling the describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json. 3. For each RDS instance, inspected the `BackupRetentionPeriod` to determine if it is greater than or equal to 14 days.
Conclusion	Fail
Comments	Exceptions Noted. 1 RDS instance(s) do not retain backups for at least 14 days.

Region	DB Instance	Result	Comments
us-east-1	prod-db-1	Fail	Retention is 7 days

Test ID	ebs_tags
Test Description	EBS volumes must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained the list of EBS volumes by calling describe_volumes() boto3 command. 2. Saved the list of volumes in the audit evidence folder (ec2/[region_name]/volumes.json). 3. For each volume, obtained its tags from the 'Tags' attribute. 4. Inspected each EBS volume to determine if the following tag keys exist and have non-empty values: ['owner', 'description', 'classification']
Conclusion	Fail
Comments	Exceptions Noted. 1 EBS volume(s) are missing required tags or have empty values.

Region	Volume ID	Result	Comments
us-east-1	vol-0fad5b6c1c9198060	Fail	Missing tags: ['owner', 'description', 'classification'].

Test ID	ec2_tags
Test Description	EC2 instances must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained the list of EC2 instances by calling describe_instances() boto3 command. 2. For each in-scope AWS region, saved the list of EC2 instances: ec2/[region_name]/instances.json 3. For each EC2 instance, reviewed the 'Tags' to determine if the following tag keys exist and have non-empty values: ['owner', 'description', 'classification']
Conclusion	Fail
Comments	Exceptions Noted. 1 EC2 instance(s) are missing required tags or have empty values.

Region	Instance ID	Result	Comments
us-east-1	i-00ad64923065104d8	Fail	Missing tags: ['owner', 'description', 'classification'].

Test ID	ec2_security_group_tags
Test Description	EC2 security groups have required tags applied and tag values are not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained a list of EC2 security groups by calling describe_security_groups() boto3 command. 2. For each in-scope region, saved the list of security groups: ec2/[region]/security_groups.json 3. Inspected each security group's 'Tags' attribute to determine if the following tag keys exist and have non-empty values: ['Owner', 'Description', 'ReviewedBy', 'LastReviewedDate']
Conclusion	Fail
Comments	Exceptions Noted. 4 security group(s) are missing required tags or have empty values.

Region	Security Group ID	Result	Comments
us-east-1	sg-0d3a9f0b108d87954	Fail	Missing tags: ['ReviewedBy', 'LastReviewedDate'].
us-east-1	sg-0c8fdfe934b59f4fa	Fail	Missing tags: ['Owner', 'Description', 'ReviewedBy', 'LastReviewedDate'].
us-east-1	sg-0f1c8a3d07e49bb15	Fail	Missing tags: ['Owner', 'Description', 'ReviewedBy', 'LastReviewedDate'].
us-east-2	sg-0281bd23cf2b7ec63	Fail	Missing tags: ['Owner', 'Description', 'ReviewedBy', 'LastReviewedDate'].

Test ID	s3_secure_transport
Test Description	S3 buckets are configured to deny unencrypted data in-transit.
Risk Rating	Informational
Test Procedures	1. Obtained a list of S3 buckets by calling the list_buckets() boto3 command. 2. Saved the list of buckets: s3/buckets.json. 3. For each bucket, obtained the bucket policy by calling the get_bucket_policy() boto3 command. 4. For each bucket, saved the bucket policy: s3/buckets/[bucket_name]/bucket_policy.json. 5. For each bucket, inspected the bucket policy to determine if a statement exists that denies requests when aws:SecureTransport is false.
Conclusion	Fail
Comments	Exceptions Noted. 5 S3 bucket(s) do not enforce secure transport (HTTPS).

Bucket Name	Result	Comments
aws-cloudtrail-logs-798094440826-aa81e64c	Fail	Unable to parse bucket policy.
bad-bucket-54321	Fail	Unable to parse bucket policy.
config-bucket-798094440826	Fail	Unable to parse bucket policy.
itauditguy	Fail	Unable to parse bucket policy.
taylor-swift-secret-vault	Fail	Unable to parse bucket policy.

Test ID	ebs_default_encryption
Test Description	EBS volumes must have default encryption enabled in each region.
Risk Rating	Informational
Test Procedures	1. For each in-scope region, obtained the EBS default encryption settings by calling get_ebs_encryption_by_default() boto3 command. 2. For each in-scope region, saved the EBS default encryption settings: ec2/[region_name]/default_ebs_encryption.json. 3. Inspected the configuration for each region to determine if 'EbsEncryptionByDefault' is set to True.
Conclusion	Fail
Comments	Exceptions Noted. 2 region(s) do not have EBS default encryption enabled.

Region	Result	Comments
us-east-1	Fail	EBS default encryption is not enabled in this region.
us-east-2	Fail	EBS default encryption is not enabled in this region.

Test ID	iam_root_mfa
Test Description	Root account has MFA enabled.
Risk Rating	High
Test Procedures	1. Obtained the AWS account summary by calling the get_account_summary() boto3 command. 2. Saved the account summary: iam/account_summary.json 3. Inspected the account summary to determine if 'AccountMFAEnabled' is set to 1.
Conclusion	Pass

Test ID	iam_root_access_key
Test Description	Root account does not have any active access keys.
Risk Rating	High
Test Procedures	1. Obtained the AWS account summary by calling the get_account_summary() boto3 command. 2. Saved the account summary: iam/account_summary.json. 3. Inspected the account summary to determine if 'AccountAccessKeysPresent' is set to 0.
Conclusion	Pass

Test ID	s3_public_access
Test Description	S3 buckets are configured to block public access.
Risk Rating	High
Test Procedures	1. Obtained a list of S3 buckets by calling the list_buckets() boto3 command. 2. Saved the list of buckets: s3/buckets.json. 3. For each bucket, obtained the public access block settings by calling the get_public_access_block() boto3 command. 4. For each bucket, saved the public access block settings: s3/buckets/[bucket_name]/public_access_block.json. 5. For each bucket, inspected the public access block settings to determine if they comply with the test attribute(s) below.
Test Attributes	• BlockPublicAcls, IgnorePublicAcls, BlockPublicPolicy, and RestrictPublicBuckets are set to true.
Conclusion	Pass

Bucket Name	Result	Comments
bad-bucket-54321	Excluded	Sample is excluded. See config.json
itauditguy	Excluded	Sample is excluded. See config.json
taylor-swift-secret-vault	Excluded	Sample is excluded. See config.json
aws-cloudtrail-logs-798094440826-aa81e64c	Pass	
config-bucket-798094440826	Pass	

Test ID	rds_public_access
Test Description	RDS instances are not publicly accessible.
Risk Rating	High
Test Procedures	1. For each in-scope region, obtained a list of RDS instances by calling the describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json) 3. For each RDS instance, inspected the 'PubliclyAccessible' setting to determine if it was set to 'false'.
Conclusion	Pass

Region	DB Instance	Result	Comments
us-east-1	prod-db-1	Pass	

Test ID	cloudtrail_multi_region
Test Description	At least one multi-region CloudTrail trail has logging enabled.
Risk Rating	High
Test Procedures	1. Obtained a list of CloudTrail trails by calling the describe_trails() boto3 command. 2. Saved the list of CloudTrail trails: cloudtrail/trails.json. 3. For each CloudTrail trail, inspected the trail configuration to determine whether 'IsMultiRegionTrail' is set to 'true'. 4. For each multi-region trail, obtained the trail status by calling the get_trail_status() boto3 command. 5. For each multi-region trail, saved the trail status: cloudtrail/trails/[trail_name]/trail_status.json. 6. Inspected the trail configuration and status to determine if at least one trail complies with the test attribute(s) defined below.
Test Attributes	• At least one trail must have IsMultiRegionTrail = true and IsLogging = true.
Conclusion	Pass

Test ID	cloudtrail_s3_bucket_protection
Test Description	CloudTrail S3 buckets are configured to block public access.
Risk Rating	High
Test Procedures	1. Obtained a list of CloudTrails by calling the describe_trails() boto3 command. 2. Saved the list of trails in the audit evidence folder (cloudtrail/trails.json). 3. For each trail, obtained the S3 bucket name and checked the bucket's public access block settings using get_public_access_block() boto3 command. 4. Saved the public access block settings for each bucket in the audit evidence folder (s3/buckets/[bucket_name]/public_access_block.json). 5. Inspected the public access block settings for each S3 bucket containing CloudTrail logs to determine if they comply with the test attribute(s) below.
Test Attributes	• CloudTrail S3 buckets must block public access (BlockPublicAcls, IgnorePublicAcls, BlockPublicPolicy, and RestrictPublicBuckets = True).
Conclusion	Pass

Trail Name	Bucket Name	Result	Comments
management-events	aws-cloudtrail-logs-798094440826-aa81e64c	Pass	

Test ID	cloudtrail_logging_recent_stops
Test Description	CloudTrail logging has not been stopped in the last 365 days.
Risk Rating	High
Test Procedures	1. Obtained a list of CloudTrails by calling the describe_trails() boto3 command. 2. Saved the list of trails in the audit evidence folder (cloudtrail/trails.json). 3. For each trail, called get_trail_status() to check IsLogging and StopLoggingTime. 4. Saved each trail's status in the audit evidence folder (cloudtrail/trails/[trail_name]/trail_status.json). 5. Inspected the 'TimeLoggingStopped' variable to determine if logging has been stopped in the last 365 days.
Test Attributes	• 'TimeLoggingStopped' must be empty OR is more than 365 days ago.
Conclusion	Pass

Trail Name	Is Logging	Last Stop Time	Result	Comments
management-events	Pass			

Test ID	iam_password_policy
Test Description	IAM passwords must comply with the organizations password complexity requirements.
Risk Rating	Medium
Test Procedures	1. Obtained the IAM password configuration by calling the get_account_password_policy() boto3 command. 2. Saved the AWS password policy: iam/password_policy.json. 3. Inspected the password configuration to determine if they comply with the test attribute(s) defined below.
Test Attributes	• MinimumPasswordLength must be >= 12. • At least 4 complexity types (RequireSymbols, RequireNumbers, RequireUppercaseCharacters, and RequireLowercaseCharacters) are set to True. • PasswordReusePrevention must be >= 24.
Conclusion	Pass

Test ID	s3_encryption
Test Description	S3 buckets are encrypted at rest.
Risk Rating	Medium
Test Procedures	1. Obtained a list of S3 buckets by calling the list_buckets() boto3 command. 2. Saved the list of buckets: s3/buckets.json. 3. For each S3 bucket, obtained the encryption settings by calling the get_bucket_encryption() boto3 command. 4. For each S3 bucket, saved the encryption settings: s3/buckets/[bucket_name]/encryption.json. 5. For each S3 bucket, inspected the encryption settings to determine if they comply with the test attribute(s) below.
Test Attributes	• ServerSideEncryptionConfiguration is present in encryption.json.
Conclusion	Pass

Bucket Name	Result	Comments
aws-cloudtrail-logs-798094440826-aa81e64c	Pass	
bad-bucket-54321	Pass	
config-bucket-798094440826	Pass	
itauditguy	Pass	
taylor-swift-secret-vault	Pass	

Test ID	rds_encryption
Test Description	RDS instances are encrypted at rest.
Risk Rating	Medium
Test Procedures	1. For each in-scope region, obtained a list of RDS instances by calling the describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json. 3. For each RDS instance, inspected the `StorageEncrypted` setting to determine if it was set to `true`.
Conclusion	Pass

Region	DB Instance	Result	Comments
us-east-1	prod-db-1	Pass	

Test ID	rds_deletion_protection
Test Description	RDS instances have deletion protection enabled at the cluster or instance level.
Risk Rating	Medium
Test Procedures	1. For each in-scope region, obtained a list of RDS instances and RDS clusters using describe_db_instances() and describe_db_clusters() boto3 commands. 2. Saved the list of RDS instances: rds/[region_name]/db_instances.json and DB clusters: rds/[region_name]/db_clusters.json. 3. Inspected each RDS instance to determine if 'DeletionProtection' was set to 'true' at the instance or cluster level.
Conclusion	Pass

Region	DB Instance	Result	Comments
us-east-1	prod-db-1	Pass	

Test ID	wafv2_enabled
Test Description	WAF is enabled on Application Load Balancers and API Gateways.
Risk Rating	Medium
Test Procedures	1. For each in-scope region, obtained a list of Web ACLs by calling list_web_acls() boto3 command. 2. Saved the list of Web ACLs (wafv2/[region]/web_acls.json). 3. For each Web ACL, obtained a list of associated resources by calling the list_resources_for_web_acl() boto3 command. 4. By default the list_resources_for_web_acl() only provide a list of Application Load Balancers. 5. Saved the Application Load Balancers associated with the ACL (wafv2/[region]/[web_acl_name]/resources.json). 6. Re-ran the list_resources_for_web_acl() boto3 command to get the associated API Gateways. 7. For each in-scope region, obtained a list of Application Load Balancers using describe_load_balancers() boto3 command. 8. Saved the list of load balancers in the audit evidence folder (elbv2/[region_name]/load_balancers.json). 9. For each load balancer, checked if Load Balancer ARN was associated with a Web ACL. 10. For each in-scope region, obtained a list of API Gateways using get_rest_apis() boto3 command. 11. Saved the list of API Gateways in the audit evidence folder (apigateway/[region_name]/rest_apis.json). 12. For each API gateway to check if it was associated with a Web ACL.
Conclusion	Pass

Region	Resource Type	Resource ID	Result	Comments
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Test ID	rds_auto_minor_version_upgrade
Test Description	RDS instances have automatic minor version upgrades enabled.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained a list of DB instances by calling the describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json. 3. For each RDS instance, inspected the 'AutoMinorVersionUpgrade' setting to determine if it was set to 'true'.
Conclusion	Pass

Region	DB Instance	Result	Comments
us-east-1	prod-db-1	Pass	

Test ID	rds_tags
Test Description	RDS instances must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained a list of RDS instances by calling describe_db_instances() boto3 command. 2. For each in-scope region, saved the list of RDS instances: rds/[region_name]/db_instances.json). 3. For each RDS instance, reviewed the `TagList` to determine if the following tag keys exist and have non-empty values: ['owner', 'description', 'classification']
Conclusion	Pass

Region	DB Instance	Result	Comments
us-east-1	prod-db-1	Pass	

Test ID	lambda_tags
Test Description	Lambda functions must have required tags applied and tag values must not be empty.
Risk Rating	Low
Test Procedures	1. For each in-scope region, obtained the list of Lambda functions by calling list_functions() boto3 command. 2. Saved the list of functions in the audit evidence folder (lambda/[region_name]/functions.json). 3. For each function, obtained its tags using list_tags() boto3 command. 4. Saved the tags for each function in the audit evidence folder (lambda/[region_name]/functions/[function_name]/tags.json). 5. Inspected each Lambda function to determine if the following tag keys exist and have non-empty values: ['owner', 'description', 'classification']
Conclusion	Pass

Region	Function Name	Result	Comments
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